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Prop divisor cero no unidad

Proposición 1. *Sea R un anillo, entonces $\{divisores\ de\ 0\} \cap \{unidades\} = \varnothing$.*

***Demostración*:** Supongamos por contradicción que $\exists a \in R : a$ es invertible y divisor de 0. Entonces, $\exists b \in R \setminus \{0\} : ab = 0$ y $\exists a^{-1} \in R : aa^{-1} = 1$. Por lo tanto, tenemos que

$$a \cdot b = 0 \implies a^{-1} \cdot a \cdot b = a^{-1} \cdot 0 \implies 1 \cdot b = 0 \implies b = 0. \quad \longrightarrow\!\longleftarrow$$

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